Recurrence in Dynamical Systems as an Engine for Additive Number Theory: A Computational Companion

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Abstract

A remarkable chain of ideas, initiated by Poincaré and brought to maturity by Furstenberg, shows that theorems about orbits of dynamical systems returning near their starting points are equivalent to theorems about arithmetic structure inside dense sets of integers. The chain runs from Poincaré recurrence and Kac's return-time lemma, through topological multiple recurrence (equivalent to van der Waerden's theorem), to measure-theoretic multiple recurrence and the Furstenberg correspondence principle (equivalent to Szemerédi's theorem), and on to polynomial recurrence (the Furstenberg–Sárközy theorem on square differences). This expository paper presents the chain alongside an open-source Python package, recurrence-arithmetic, in which every object appearing in the proofs—invariant measures, correlation integrals, orbit closures, return times—is given an exact finite analogue that can be computed and tested. We report numerical verifications of Kac's lemma to four decimal places, exact machine certifications of the van der Waerden numbers $W(2,3) = 9$ and $W(3,3) = 27$, empirical multiple-recurrence correlations exhibiting explicit four-term progressions in structured sparse sets, and Weyl-sum computations quantifying the polynomial recurrence underlying Sárközy's theorem. The aim is pedagogical: to make the dictionary between ergodic theory and additive combinatorics tangible.

1 Introduction

Szemerédi's theorem [5] states that every set of integers of positive upper density contains arbitrarily long arithmetic progressions. Two years after Szemerédi's combinatorial proof, Furstenberg [6] gave a second proof of an entirely different character: he showed the theorem is equivalent to a statement about measure-preserving dynamical systems—that any set of positive measure returns to itself along arithmetic progressions of times—and then proved that dynamical statement by structural methods in ergodic theory. The translation device, now called the Furstenberg correspondence principle, opened a two-way street between ergodic theory and additive combinatorics that has carried traffic ever since, including the Bergelson–Leibman polynomial Szemerédi theorem [9] and, in spirit, the Green–Tao theorem on progressions in the primes [10].

The proofs in this circle are infinitary: they invoke weak-$*$ limits, invariant measures on compact spaces, and transfinite structure theorems. No finite computation

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can replace them. Nevertheless, every object the proofs manipulate has an exact finite shadow. The measure of a cylinder set is a density; a multiple-recurrence integral is a correlation count; an orbit closure is a set of shifted windows; a return time is a gap between visits. These shadows can be computed, and watching them behave as the theorems predict is, in our experience, the fastest way to internalize why the correspondence principle works.

This paper does two things. Sections 2–5 give a self-contained exposition of the recurrence chain

$$\text{Poincaré} \longrightarrow \text{Kac} \longrightarrow \text{van der Waerden} \longrightarrow \text{Szemerédi} \longrightarrow \text{Sárközy},$$

stating each dynamical theorem next to its combinatorial twin. Section 6 describes the accompanying software package and reports the numerical experiments; Section 7 discusses limitations and extensions.

2 Poincaré recurrence and Kac's lemma

Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system: X a probability space and $T\colon X \to X$ a measurable map with $\mu(T^{-1}A) = \mu(A)$ for all measurable A.

Theorem 2.1 (Poincaré, 1890 [1]). If $\mu(A) > 0$, then μ-almost every $x \in A$ satisfies $T^n x \in A$ for infinitely many $n \geq 1$.

The proof is three lines of pigeonhole, yet the statement is the seed of everything below. Kac made it quantitative for ergodic systems.

Theorem 2.2 (Kac, 1947 [4]). Suppose (X, μ, T) is ergodic and $\mu(A) > 0$. Let $r_A(x) = \min\{n \geq 1 : T^n x \in A\}$ be the first-return time. Then

$$\int_A r_A \, d\mu = 1, \qquad \text{equivalently} \qquad \mathbb{E}_{\mu_A}[r_A] = \frac{1}{\mu(A)},$$

where μ_A is the normalized restriction of μ to A.

Our first computational touchstone: for the golden rotation $T(x) = x + \alpha \bmod 1$ with $\alpha = (\sqrt{5}-1)/2$ and $A = [0, 0.1)$, Theorem 2.2 predicts mean return time exactly 10. A single orbit of length 2×10^5 yields empirical mean 9.9999 (Section 6). The return-time sequence is not constant—for the golden rotation it takes the three values $5, 8, 13$ (consecutive Fibonacci numbers, by the three-distance theorem)—but its average obeys Kac exactly.

3 Topological multiple recurrence and van der Waerden

Theorem 3.1 (van der Waerden, 1927 [3]). For all $r, k \in \mathbb{N}$ there exists $N = W(r, k)$ such that every r-colouring of $\{1, \ldots, N\}$ contains a monochromatic k-term arithmetic progression.

Furstenberg and Weiss [8] derived this from a purely dynamical statement.

Theorem 3.2 (Topological multiple recurrence). Let T be a homeomorphism of a compact metric space (X, d). For every $k \in \mathbb{N}$ and $\varepsilon > 0$ there exist $x \in X$ and $n \geq 1$ with

$$d(T^{jn}x, x) < \varepsilon \qquad \text{for } j = 1, \ldots, k-1.$$

Dictionary from Theorem 3.2 to van der Waerden. An r-colouring $c\colon \mathbb{Z} \to \{1, \dots, r\}$ is a point of the compact metric space $\Omega = \{1, \dots, r\}^{\mathbb{Z}}$, where two colourings are 2^{-m}-close when they agree on the window $[-m, m]$. Let X be the orbit closure of c under the shift σ. Applying Theorem 3.2 with $\varepsilon < 1$ produces $x \in X$ and n such that $x, \sigma^n x, \dots, \sigma^{(k-1)n} x$ all agree at the origin; since x is a limit of shifts of c, this agreement pulls back to integers $m, m+n, \dots, m+(k-1)n$ receiving the same colour. □

Two computational remarks. First, the theorem is finitary enough that small van der Waerden numbers can be certified by exhaustive search: our package's backtracking procedure confirms $W(2,3) = 9$ and $W(3,3) = 27$, in each case exhibiting an AP-free witness colouring of $\{1, \dots, N-1\}$ and proving by exhaustion that none exists for $\{1, \dots, N\}$. Second, the dynamical proof delivers strictly more than a monochromatic progression: taking $\varepsilon = 2^{-m}$ in Theorem 3.2 yields shifted colourings agreeing on windows of length m. Our orbit_closure_recurrence routine locates this stronger agreement in concrete colourings, making the gap between the combinatorial statement and its dynamical strengthening visible.

4 The correspondence principle and Szemerédi's theorem

For $S \subseteq \mathbb{N}$ write $\overline{\mathrm{d}}(S) = \limsup_{N \to \infty} \frac{1}{N} |S \cap [0, N)|$ for the upper density.

Theorem 4.1 (Szemerédi, 1975 [5]). If $\overline{\mathrm{d}}(S) > 0$ then S contains a k-term arithmetic progression for every k.

Theorem 4.2 (Furstenberg multiple recurrence, 1977 [6]). Let $(X, \mathcal{B}, \mu, T)$ be measure preserving and $\mu(A) > 0$. For every $k \geq 1$ there exists $n \geq 1$ with

$$\mu\big(A \cap T^{-n}A \cap T^{-2n}A \cap \dots \cap T^{-(k-1)n}A\big) > 0.$$

Proposition 4.3 (Correspondence principle). Theorem 4.2 implies Szemerédi's theorem.

Sketch. Let $\omega \in \{0,1\}^{\mathbb{N}}$ be the indicator sequence of S and let $A = \{\xi : \xi_0 = 1\}$ be a cylinder in the shift space. Choose $N_j \to \infty$ realizing the $\limsup$ in $\overline{\mathrm{d}}(S)$ and let μ be a weak-$*$ limit point of the empirical measures $\frac{1}{N_j} \sum_{i<N_j} \delta_{\sigma^i \omega}$. Then μ is shift invariant, $\mu(A) \geq \overline{\mathrm{d}}(S) > 0$, and for the closed-and-open set $A \cap \sigma^{-n} A \cap \dots \cap \sigma^{-(k-1)n} A$ the μ-measure is dominated by the upper density of $\{m : \ m, m+n, \dots, m+(k-1)n \in S\}$. Positivity of the former (Theorem 4.2) forces the latter to be nonempty: a k-term progression in S. □

The finite shadow of the measure in Proposition 4.3 is the correlation

$$c_k(n) \ = \ \frac{1}{N} \# \{ \, m : \ m, \ m+n, \ \dots, \ m+(k-1)n \in S \cap [0, N) \, \}, \tag{1}$$

which is exactly the empirical value of $\mu\big(A \cap T^{-n}A \cap \dots \cap T^{-(k-1)n}A\big)$ along the orbit of ω. Multiple recurrence asserts that some n makes (1) positive; Szemerédi asserts S contains a k-term progression. In the finite model these are the same sentence, and the package computes both sides of the identity.

5 Polynomial recurrence and the Furstenberg–Sárközy theorem

Recurrence persists along much sparser sets of times than all of $\mathbb{N}$. The prototype is quadratic.

Theorem 5.1 (Furstenberg [6]; Sárközy [7]). If $\overline{\mathrm{d}}(S) > 0$, there exist $a, b \in S$ and $n \geq 1$ with $a - b = n^2$. Dynamically: $\mu(A) > 0$ implies $\mu(A \cap T^{-n^2} A) > 0$ for some $n \geq 1$.

The dynamical input is equidistribution of $n^2\alpha \bmod 1$, which is generated by the affine skew product on the torus $\mathbb{T}^2$,

$$T(x, y) = (x + \alpha,\ y + 2x + \alpha) \pmod 1, \qquad T^n(0, 0) = (n\alpha,\ n^2\alpha), \tag{2}$$

a uniquely ergodic system for irrational α. Unique ergodicity forces the orbit of the origin to be dense; in particular $n^2\alpha$ returns arbitrarily close to 0—polynomial recurrence. Weyl's criterion [2] converts equidistribution into decay of the exponential sums $\frac{1}{N}\sum_{n<N} \mathrm{e}^{2\pi i n^2 \alpha}$, which the package computes directly (Table 1).

6 The computational companion

6.1 Package design

The package recurrence-arithmetic (Python, NumPy only, MIT licence) mirrors the mathematical chain module by module: systems implements the circle rotation, doubling map, the skew product (2), and shift systems built from integer sets; poincare computes visit times and Kac averages; multiple computes the correlations (1) vectorised over m and locates explicit progressions; correspondence packages the finite-N correspondence dictionary; vanderwaerden contains the exact backtracking decision procedure and the orbit-closure recurrence search; and equidistribution computes Weyl sums, star discrepancy, and square-difference witnesses. Eleven unit tests bind the code to the theorems: for instance, one test asserts that the correlation (1) of the even numbers at $(k, n) = (3, 2)$ equals the exact count $8/20$, and another certifies $W(2, 3) = 9$ in both directions.

The heart of the correspondence module fits in one line of NumPy: for a $0/1$ word w of length N,

```
prod = w[: N-(k-1)*n].copy()
for j in range(1, k):
    prod &= w[j*n : N-(k-1-j)*n] # A & T^{-jn} A
c_k_n = prod.sum() / N # empirical multiple-recurrence measure
```

which computes (1) in $O(kN)$ time per gap n.

6.2 Experiments

Kac's lemma. Golden rotation, $A = [0, 0.1)$, orbit length 2×10^5 from $x_0 = 0.05$: the orbit visits A exactly $20{,}000$ times; the first return gaps are $13, 8, 13, 8, 5, 8, 13, \ldots$ (Fibonacci values); the empirical mean return time is 9.9999 against the Kac prediction 10.0000.

Van der Waerden. The backtracking procedure certifies $W(2,3) = 9$ (witness AP-free colouring of $\{1,\ldots,8\}$: 00110011) and $W(3,3) = 27$, agreeing with the classical values [3]. On a uniformly random 3-colouring of length 5000, the orbit-closure search finds shifts $\sigma^m, \sigma^{m+n}, \sigma^{m+2n}$ agreeing on windows of length 3—the strengthened recurrence of Theorem 3.2 with $\varepsilon = 2^{-3}$.

Szemerédi via correspondence. Let S be the integers in $[0, 60000)$ with no digit 3 in base 4, thinned by independent fair coin flips; the realized set has 3343 elements and density 0.0557. For $k = 4$ the correlation profile $c_4(n)$ is positive at many gaps, and an explicit progression is extracted:

$$\{\,1,\ 2049,\ 4097,\ 6145\,\} \subset S, \qquad \text{common difference } 2048 = 2 \cdot 4^5.$$

The power-of-four structure of the gap is forced by the digit condition, which makes S nearly invariant under suitable shifts: the recurrence search discovers the set's hidden structure unaided. This is a miniature of the structure-versus-randomness dichotomy that organizes all known proofs of Szemerédi's theorem [11].

Polynomial recurrence and square differences. Table 1 shows the decay of the normalized Weyl sums for $\alpha = \sqrt{2} - 1$, confirming equidistribution of both $n\alpha$ and $n^2\alpha$; the star discrepancy of $(n^2\alpha)_{n<10^5}$ is 0.00279. The first times with $\|n^2\alpha\| < 10^{-4}$ are $n = 6201, 6227, 7621, \ldots$—explicit polynomial recurrence. On the arithmetic side, a random set of density 5% in $[0, 20000)$ yields a square difference immediately, and the empirical profile $n \mapsto \mu(A \cap T^{-n^2} A)$ is uniformly positive over $1 \le n \le 10$, as Theorem 5.1 predicts for sets with no structure conspiring against squares.

N	$\lvert\frac{1}{N}\sum_{n<N} \mathrm{e}^{2\pi i n\alpha}\rvert$	$\lvert\frac{1}{N}\sum_{n<N} \mathrm{e}^{2\pi i n^2\alpha}\rvert$
10^3	0.00064	0.01834
10^4	0.00004	0.00979
10^5	0.00001	0.00134

Table 1: Decay of normalized Weyl sums for $\alpha = \sqrt{2} - 1$. Linear sums decay like $O(1/N)$ for badly approximable α; quadratic sums decay more slowly, reflecting the weaker (but still sufficient) equidistribution driving polynomial recurrence.

7 Discussion

What the computations do and do not show. No finite experiment proves an infinitary theorem, and we claim none does. What the experiments provide is (i) verification that the finite analogues behave exactly as the theorems dictate, to high precision (Kac) or with exact certificates (van der Waerden); and (ii) visibility: the correspondence principle, usually met as a compactness argument, becomes a dictionary one can evaluate entry by entry. In the finite model, the identity between "$\mu(A \cap T^{-n} A \cap \cdots) > 0$" and "$S$ contains a progression" is not an implication to be proved but a single quantity computed two ways.

Extensions. Natural next modules include the Bergelson–Leibman polynomial Szemerédi theorem [9] (replace the gaps $n, 2n, \ldots$ in (1) by arbitrary integer polynomials vanishing at zero), IP-recurrence and Hindman-type phenomena, and quantitative comparisons of $c_k(n)$ against the bounds emerging from the recent quasi-polynomial progress on Szemerédi-type problems. The package's finite-correlation framework accommodates all of these with small changes.

Reproducibility. All figures in Section 6 are produced by the four example scripts shipped with the repository; the test suite runs in under a second, and the full experiment suite in under two minutes on a laptop.

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